Dr Bradley A. A. Martin

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Scholar	https://scholar.google.com/citations?hl=en&user=syAS4SgAAAAJ
GitHub	https://github.com/Neutrino155
MACEField	https://github.com/mdi-group/mace-field

Research Profile

Theoretical physicist and materials machine-learning researcher working on physics-informed ML interatomic potentials, electric response, charge density, source-conjugate generating functionals, and field-driven atomistic simulation.

Research Themes

- Field-aware MLIPs for dielectric and ferroelectric response.
- Source-conjugate learning: J -> F(R,J) -> X = dF/dJ.
- Long-range electrostatics, charged defects, density-aware models, and electron-phonon physics.
- Path-integral and variational methods for polaron transport.

Selected Publications

General Learning of the Electric Response of Inorganic Materials
B. A. A. Martin, A. M. Ganose, V. Kapil, T. Li, and K. T. Butler
PRX Intelligence 1, 013006 (2026). DOI: 10.1103/b116-xy8k. arXiv:2508.17870.

Six Open Questions in Machine-Learned Interatomic Potential Foundation Models
I. Creed et al., including B. A. A. Martin
arXiv:2606.07327 (2026).

Discovery and recovery of crystalline materials with property-conditioned transformers
C. Bone, M. Walker, B. A. A. Martin, et al.
arXiv:2511.21299 (2025; v3 2026).

Accelerating molecular dynamics by going with the flow
A. Y. Ismail, B. A. A. Martin, and K. T. Butler
Nature Machine Intelligence 7, 1598-1599 (2025). DOI: 10.1038/s42256-025-01129-0.

Predicting polaron mobility in organic semiconductors with the Feynman variational approach
B. A. A. Martin and J. M. Frost
arXiv:2207.06846 (2022; revised 2024).

Selected Projects and Software

- MACEField: electric-field-aware MACE extension for finite-field simulations and response-property prediction.
- Source-conjugate MLIPs: generating-functional models whose derivatives expose physical response channels.
- Path-integral polarons: variational path-integral methods for charge-carrier mobility and electron-phonon coupling.

Talks, Service, and Collaboration

- Invited CECAM-US-CENTRAL talk on AI methods for materials simulation and discovery.
- MRS Boston presentation on machine learning for materials response.
- PSDI collaboration on research software and physical-sciences data infrastructure.
- MSc project development around finite-field MLIPs and ferroelectric switching pathways.

Note	Concise web CV. ORCID and Google Scholar are linked for the current full publication record.